

Internet of Things (IoT)

Lecture 5: IoT Protocols

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IoT Course Topics

1. Introduction to IoT
2. IoT Architecture & Network Components
3. Smart Objects & IoT Devices
4. Communication Technologies in IoT
5. **IoT Protocol Stacks**
6. IoT Application Protocols & Data Formats
7. IoT Security Principles
8. Cloud & Fog Computing in IoT
9. Data Analytics in IoT
10. IoT Platforms & Middleware
11. Designing IoT Solutions



IoT Protocols



IoT Protocols

- The IoT system can only perform and transfer information when it's in online mode, which means the devices in IoT must be safely connected to communication networks.

➤ The question that arises is how these devices can connect and communicate with each other.?



✓ The answer lies in Protocols.

➤ IoT protocols are essential for enabling seamless and efficient communication between Internet of Things (IoT) devices, applications, and services. They serve as the rules and standards that govern how devices in an IoT ecosystem exchange data and interact with each other.

Why do we need IoT Protocols?

- The ability to interact with each other and resolve common problems is what separates IoT devices from traditional computers.
- These interactions are only possible if there is a medium or means of communication in the IoT ecosystem.
- The IoT protocols are thus a common “language” that allows devices to interact with other IoT devices.
- The IoT protocols lay down standards that are adopted in every IoT ecosystem for proper functioning and to avoid security threats.
- There is no universal protocol that helps in communicating with all IoT devices yet.

Why do we need IoT Protocols?

Interoperability: In IoT, devices come from various manufacturers, and they may use different communication technologies and data formats. IoT protocols ensure that devices from different vendors can communicate and work together seamlessly, fostering interoperability and creating a unified IoT ecosystem.

Efficient Communication: IoT protocols are designed to optimize data transmission and reduce unnecessary overhead, leading to efficient use of resources like power and bandwidth. This efficiency is crucial, especially for IoT devices with limited processing capabilities and battery life.

Security: IoT devices often handle sensitive data and are vulnerable to cybersecurity threats. IoT protocols incorporate security mechanisms like encryption, authentication, and data integrity checks to ensure secure communication and protect against unauthorized access.

Scalability: IoT deployments can involve a large number of devices spread across wide areas. IoT protocols are designed to handle scalability, allowing the network to accommodate a growing number of devices without compromising performance.

Why do we need IoT Protocols?

Low Power Consumption: Many IoT devices are battery-powered or operate in energy-constrained environments. IoT protocols, especially those designed for low-power communication, help extend the battery life of devices and minimize energy consumption.

Real-time Communication: Some IoT applications, such as industrial automation and healthcare monitoring, require real-time data exchange. IoT protocols cater to such scenarios and ensure timely delivery of critical information.

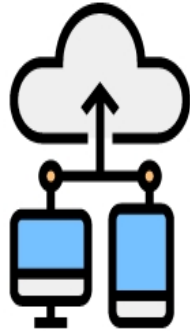
Standardization: IoT protocols are standardized to ensure consistency and compatibility across different implementations. This standardization facilitates the development of a diverse range of IoT devices and applications while ensuring they can work together smoothly.

Data Management: IoT protocols define how data is structured, transmitted, and processed, facilitating efficient data management and analysis. Proper data handling is essential for making informed decisions and deriving meaningful insights from IoT-generated data.

Types of IoT Protocols

IoT protocols are briefly classified into two types:

1. **IoT data protocols**
2. **IoT Network Protocols**



IoT Network Protocols



IoT Data Protocols

Types of IoT Protocols

IoT protocols are briefly classified into two types:

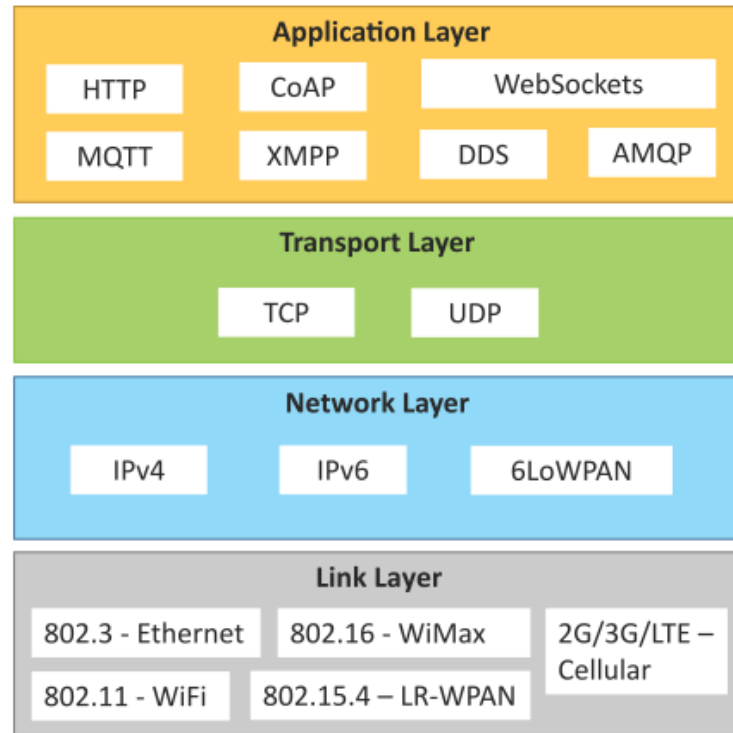
IoT data protocols are designed to handle the specific challenges of IoT systems, such as limited bandwidth, energy constraints, and varying types of devices. They provide mechanisms for data encoding, transmission, and reception.

IoT Network Protocols handle aspects like device discovery, data routing, and network management. The primary goal of IoT network protocols is to enable efficient, reliable, and secure communication over various types of networks, such as local area networks (LANs), wide area networks (WANs), or even satellite networks.

- ✓ **While** IoT data protocols are focused on the format and transmission of data between devices (e.g., MQTT, CoAP), **IoT network protocols** deal more with the physical and logical structure of the network and the paths that the data takes.

IoT Protocols

- Link Layer
 - 802.3 – Ethernet
 - 802.11 – WiFi
 - 802.16 – WiMax
 - 802.15.4 – LR-WPAN
 - 2G/3G/4G
- Network/Internet Layer
 - IPv4
 - IPv6
 - 6LoWPAN
- Transport Layer
 - TCP
 - UDP
- Application Layer
 - HTTP
 - CoAP
 - WebSocket
 - MQTT
 - XMPP
 - DDS
 - AMQP



Internet Protocols VS IoT Protocols

Current Internet Protocols

HTTP FTP,SMTP,IMAP
TCP and UDP
IPv4 and IPv6
Ethernet,Wi-Fi, GSM

Application

Transport

Networking

Data Link

Protocol Level
TCP/IP Model

Expected IOT Protocols

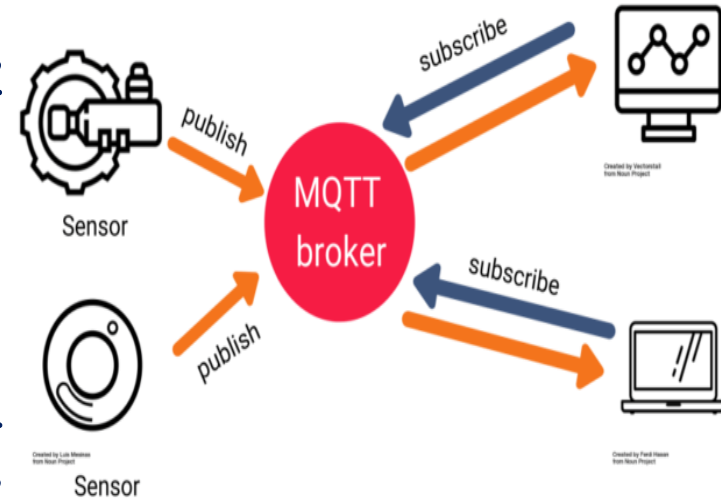
MQTT COAP,AMQP
UDP and TCP
IPv6 and IPv4
Ethernet,Wi-Fi, GSM, LTE-M, Lora, SigFox

IOT and Internet Protocols

IoT data protocols

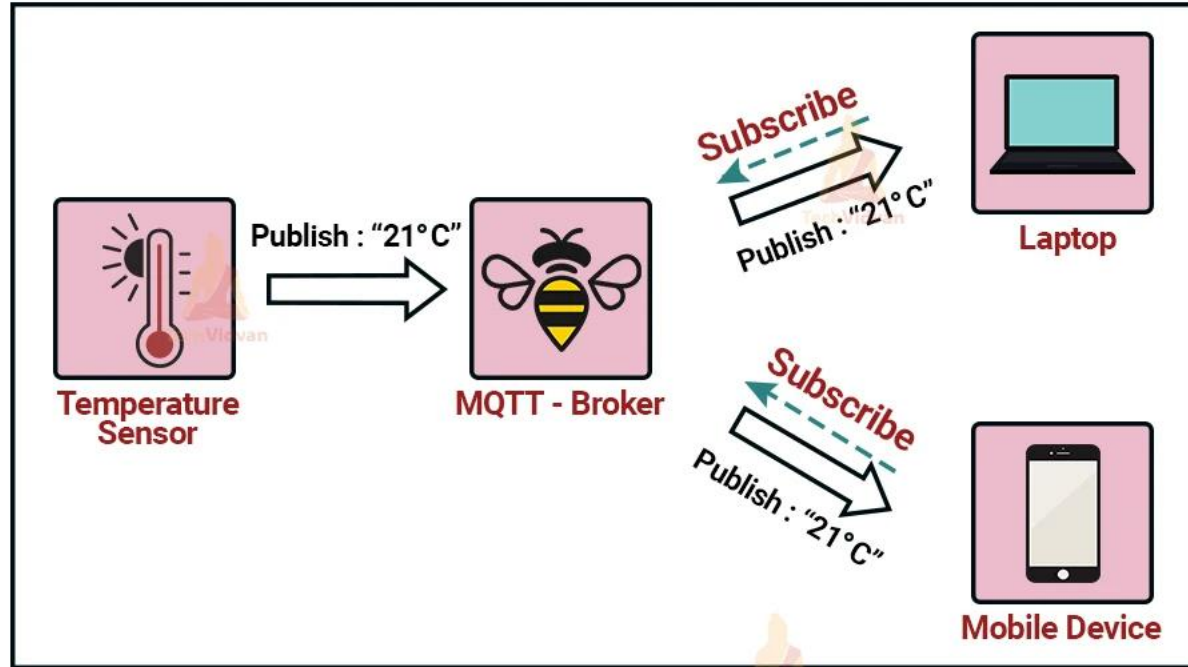
1. Message Queuing Telemetry Transport (MQTT)

- MQTT is a lightweight protocol specifically designed for sending data from sensors to middleware and applications.
- MQTT leverages TCP/IP as its underlying technology, ensuring dependable and secure data delivery. This reliability is essential for real-time and critical IoT applications.
- MQTT consists of three main components: Subscriber, Publisher, and Broker, where the publisher generates and transmits information to subscribers through a broker.
- In MQTT's basic workflow, the publisher is responsible for generating and transmitting information. The broker acts as an intermediary, managing the data and delivering it to the relevant subscribers.



Message Queuing Telemetry Transport (MQTT)

Message Queuing Telemetry Transport (MQTT)



1

Subscribe to
topic : "Temperature"

2

Publish to
topic : "Temperature"

Message Queuing Telemetry Transport (MQTT)

MQTT Clients

Ex: Sensors



Publish: "30°"
Topic: "temp"



Publish: "65%"
Topic: "moisture"



Publish: "1.8g"
Topic: "accel-x"



MQTT
Broker

MQTT Clients



Ex: Valve



Publish: "30°"

Subscribed to "temp"

Publish: "closed"
Topic: "valve"

Ex: Cloud Server



Subscribed to "accel-x"

Publish: "1.8g"

- MQTT is different to the regular client-server model. It divides clients into two groups.

Thank You